



Association of Game Use With Loneliness and Social Isolation: A Nationwide Korean Study

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Objective The aim of this article is to examine the correlation between social isolation, loneliness, and the use of online games. Conflicting conclusions have been drawn in previous studies on this topic due to small sample sizes and the confounding effects of psychiatric disorders. To address these limitations, the authors conducted a nationwide study that gives consideration to sociodemographic variables and psychiatric disorders.

Methods A total of 5,511 Koreans responded to the Loneliness and Social Isolation Scale-6, Game Overuse Screening Questionnaire, and Korean version of the Composite International Diagnostic Interview between January 2021 and March 2021. Participants were classified as non-gamer, low-risk gamer, and high-risk gamer according to their game usage. Multivariate linear regression was performed to evaluate the association of game usage with loneliness and social isolation after propensity matching controlling for sociodemographic data and presence of psychiatric disorders.

Results Low-risk gamers reported significantly lower loneliness scores (0.53 ± 1.02) compared to other gaming groups (non-risk: 0.94 ± 1.44 , high-risk: 1.02 ± 1.64). Among male participants, non-risk gamers (2.49 ± 1.51) showed lower social network scores than low-risk gamers (2.10 ± 1.11) and high-risk gamers (2.09 ± 1.31). Loneliness ($p=0.001$) was more strongly correlated with game usage than social support ($p=0.839$) or network ($p=0.055$). The relationship between loneliness and game usage was significantly stronger in non-risk ($B=0.41$) and high-risk ($B=0.44$) gamers than in low-risk gamers.

Conclusion Increased use of game does not show a linear relationship with loneliness and isolation when correcting for confounding factors including psychiatric disease. Rather, low-risk game use was associated with lower scores for loneliness and isolation. Further studies exploring other factors that affect gaming overuse, loneliness and social isolation are needed.

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Keywords Loneliness; Social isolation; Online games; Psychiatric disorders.

INTRODUCTION

Today, as direct interactions between people decrease, the phenomenon of social isolation and loneliness has become a representative keyword in the field of psychiatry.¹ Loneliness and social isolation have been found to be related to many

psychiatric disorders,² personality disorders,³ physical diseases,⁴ and metabolic disorders.⁵ While various alternatives to human contact are being suggested, online games (referred to as “games” below) are attracting attention. The game industry was previously growing rapidly,⁶ and since the COVID-19 pandemic, there has been a significant increase in individual accessibility and utilization of games.⁷

While various discussions are underway regarding increased loneliness, social isolation, and games, clear conclusions have not yet been reached with regard to the relationships among these. Existing studies^{8,9} have argued that excessive game use makes it difficult to achieve real goals and makes people socially isolated. However, other studies¹⁰ have recently shown that when the effects of existing psychiatric disorder are corrected for, protective effects of gaming such as increased so-

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cial interaction and stress relief activities reduce loneliness and social isolation. Another study¹¹ has even shown that internet gaming disorder patients perceive game-related signals as social stimuli in the brain, and this means that online game signals are linked to social signals.

This opposite conclusion could be derived from the confounding effects of psychiatric disorders, as psychiatric disorders affect increased levels of loneliness, social isolation,¹² and game use.^{13,14} Also, the recent investigations cited above are only focused on specific populations or psychiatric illnesses. Further, recent studies^{8,10} did not have sufficient demographic sample sizes for observing loneliness, social isolation, and game use, as these traits are being investigated by few large-scale censuses. Above all, more studies which correct for confounding variables including general psychiatric traits and use statistical representative values are needed.

The object of the study is to evaluate the association of game usage with loneliness and social isolation in a nationwide cohort. We applied propensity score matching analysis to reduce the differences in sociodemographic variables and presence of psychiatric disorders, which might have considerable confounding effects on patterns of game usage, loneliness, and social isolation. We hypothesized that the association of game usage with loneliness and social isolation would vary according to the level of game use.

METHODS

Data collection

The present study is based on data from the National Mental Health Survey 2021 conducted between January 1, 2021 and March 31, 2021. The Korean National Mental Health Survey is a representative sample survey on mental health of the Korean population and has been conducted every five years since 2001.¹⁵ In the 2021 study, sample households were extracted through multi-stage stratified extraction based on data from the Population and Housing Census and household register, and one adult aged 18 to 79 was randomly selected among the members of each household. Of the total 13,540 households selected, 12,588 were counted as eligible households. Of these, 5,511 completed a response, giving a 43.8% response rate.¹⁵ This study was reviewed by the Institutional Review Board of Samsung Medical Center (SMC2022-11-042). Written informed consent was obtained from all participants who were 18 years of age or older, in compliance with applicable guidelines and regulations. The study procedures adhered to the prescribed ethical standards.

Psychiatric disorders

The Korean version of the Composite International Diag-

nostic Interview 2.1 (K-CIDI) was administered to the study respondents by trained interviewers. CIDI is a standardized diagnostic tool developed for studies of cross-cultural epidemiology and psychopathology of psychiatric disorders,¹⁶ and K-CIDI is a translation of CIDI according to the socio-cultural background of Korea, and its reliability and validity were confirmed by Cho et al.¹⁷ Lifetime morbidity of depressive disorders, anxiety disorders, alcohol use disorders and nicotine dependence & withdrawal were evaluated according to the The Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV)¹⁸ diagnostic criteria. In this study, schizophrenia spectrum disorder and bipolar disorder were not included due to very low prevalence in the community.¹⁵

Loneliness and social isolation

To measure the degree of loneliness and social isolation, we used the Loneliness and Social Isolation Scale-6 (LSIS-6) model developed by Hwang et al.¹⁹ The LSIS-6 scale consists of three subscales: loneliness, social support, and social network. Each subscale consists of two questions evaluated on a scale of 0–3 points. In the subscales, higher scores indicate a higher level of loneliness, a lower level of social support, and a lower level of social network, respectively.

The scales consist of 6 questions: “Do you feel lonely?” “Do you feel left out?” (2 questions for loneliness), “Can you rely on your family or friends comfortably?” “Is there someone who can help you with your daily work?” (2 questions for social support), “How many people are you close enough with to meet privately at least once a month or contact them at least once a week? (including family, relatives, and friends),” “On average, how many minutes of the day do you spend in personal contact with your friends or family? (phone, text, online messenger, etc.)” (2 questions for social network).

Social support represents a functional part of social isolation, which includes the quantitative and behavioral aspects of interaction in social relationships.^{19–21} Social network represents a structural part of social isolation, referring to the quantitative aspect of whom subjects engage in interpersonal relationships, and the degree of connectedness between subjects and such persons.¹⁹

Online game usage

We used the Game Overuse Screening Questionnaire (GOS-Q), a self-report screening tool for game addiction,²² as a measure of game usage behavior. In this survey, the questions focused on online games. The survey consists of 30 questions, and questions are asked in the six areas of preoccupation, tolerance, craving/withdrawal, loss of control, neglect of other areas, and insight regarding game habits over the past month. Each question is a 4-point Likert tool with a total question-

naire score of 30–120, while the cut-off value for cases with a risk of game addiction is 38.5.²² We classified respondents who do not currently play games as non-gamers, those who play games with a survey score of less than 38.5 as low-risk gamers, and those who play games and with a GOS-Q survey score of 38.5 or higher as high-risk gamers.

Weighting and propensity score matching

All the data were statistically weighted with the weights applied in the National Mental Health Survey 2021 before the analysis.¹⁵

We compared the scores of LSIS-6 and its subscales among the three groups which were divided according to game use. We used propensity score matching to minimize individual differences in large-scale data and improve the representativeness of the analysis results.²³ Propensity score matching is a matching method performed using propensity scores, which

are conditional probabilities allocated to a particular (treatment) group given the observed covariates.²⁴ As the covariates for matching, sex, age group, marital status, employment status, years of education, and presence of psychiatric disorder were used.

Statistical analysis

Data was analyzed by using SAS version 9.4 (SAS institute). The chi-square tests and the Fisher's exact test were performed to check the differences in distribution among the three groups after matching, and post hoc tests were also performed to compare each pair. To confirm the effects of matching, standardized differences were calculated for each pair of the three groups, and if the absolute value of the standardized difference after matching was less than 0.2, the matching was evaluated to have been well-performed.²⁵ In this study, the differences in standardized difference in the matched variables were

Table 1. Characteristics of respondents according to game usage after propensity score matching

Variables	(1) Non-gamers (N=186)	(2) Low-risk gamers (N=175)	(3) High-risk gamers (N=208)	p	Standardized difference		
					1 vs. 2	1 vs. 2	2 vs. 3
Sex*				0.264			
Male	134 (72.2)	135 (77.5)	146 (70.3)		-0.122	0.043	0.166
Female	52 (27.8)	39 (22.5)	62 (29.7)		0.122	-0.043	-0.166
Age (yr)*				0.995			
18–29	115 (61.8)	106 (60.6)	131 (62.9)		0.023	-0.024	-0.047
30–39	48 (25.8)	46 (26.5)	52 (24.9)		-0.015	0.022	0.037
40–49	23 (12.4)	23 (12.9)	25 (12.2)		-0.014	0.005	0.020
Marital status*				0.778			
Married	40 (21.2)	36 (20.5)	38 (18.5)		0.019	0.069	0.050
Not married	147 (78.8)	139 (79.5)	170 (81.5)		-0.019	-0.069	-0.050
Employment status*				0.964			
Regular	111 (59.8)	101 (57.9)	116 (55.7)		0.039	0.085	0.046
Temporary	12 (6.7)	13 (7.2)	15 (7.2)		-0.021	-0.019	0.001
Homemaker	2 (1.2)	2 (1.2)	1 (0.6)		0.002	0.061	0.059
Student	42 (22.5)	42 (24.0)	59 (28.2)		-0.037	-0.132	-0.095
Unemployed	18 (9.9)	17 (9.7)	17 (8.4)		0.006	0.052	0.046
Education (yr)*				0.694			
<13	26 (14.2)	20 (11.2)	27 (13.0)		0.090	0.035	-0.055
≥13	160 (85.8)	155 (88.8)	181 (87.0)		-0.090	-0.035	0.055
Psychiatric disorders							
Depressive disorder†	12 (6.2)	8 (4.5)	20 (9.5)	0.145	0.076	-0.122	-0.196
Anxiety disorder‡	16 (8.6)	16 (9.1)	30 (14.5)	0.111	-0.020	-0.187	-0.167
Substance use disorder§	32 (17.3)	41 (23.7)	39 (18.7)	0.270	-0.161	-0.038	0.123
Any psychiatric disorder*	53 (28.4)	52 (29.6)	65 (31.4)	0.812	-0.026	-0.064	0.038

Data are presented with number (%). *matching variables; †major depressive disorder and dysthymia; ‡specific phobia, panic disorder, agoraphobia, social phobia, and general anxiety disorder; §alcohol use disorder and nicotine dependence & withdrawal

consistently less than 0.2, indicating that the matching was well-implemented. For the matched data, the mean and standard deviation of LSIS-6 and each of its subscales were calculated by group, and one-way analysis of variance (ANOVA) was used to determine whether there were differences in the mean among the three groups. The above ANOVA analysis was performed for all respondents, male respondents, and female respondents. In addition, we checked whether there were differences in the scores for LSIS-6 and its subscales depending on game usage group even after correcting for the effects of sociodemographic factors and the presence of psychiatric disorders through multivariate linear regression analysis. The results of analysis are expressed using the non-standardization coefficient (B), standard error (S.E.), t-statistics, and p-value for the reference of each variable. In all the above analyses, p-values of <0.05 were considered significant.

RESULTS

Characteristics of respondents according to the game usage

The characteristics of respondents before propensity score matching are shown in Supplementary Table 1. Among the respondents before matching, 2,757 (50.0%) were male and 2,754 (50.0%) were female. Age groups with less than 10% of respondents reporting game usage were excluded from the analysis, based on a previous study²⁶ which found different patterns of online game usage according to age. Thus, the age groups of 50–59 (8.6%), 60–69 (3.8%), and 70 years and older

(0.4%) were excluded. After propensity score matching (Table 1), the total number of samples was 569, with propensity scores matched in each sample for generalizability. The numbers of non-gamers, low-risk gamers, and high-risk gamers were 186 (32.7%), 175 (30.8%), and 208 (36.5%), respectively. Of the total respondents, 415 (73.1%) were male and 153 (27.0%) were female. There was no significant difference in the distribution of sex, age, marital status, employment status, and number of years of education among the three groups.

Comparison of LSIS-6 and its subscales according to the game usage

Table 2 and Figure 1 show the ANOVA results for game use groups according to LSIS-6 and its subscales. There was no significant difference in the mean total LSIS-6 score among the three groups for all respondents ($p=0.059$) and female respondents ($p=0.680$). However, when only male respondents were analyzed, the low-risk gamer group (4.86 ± 2.40) showed significantly lower total LSIS-6 scores than the other two groups (non-gamers: 5.86 ± 3.30 and high-risk gamers: 5.59 ± 3.17 , $p=0.017$). In analysis of all respondents using the three LSIS-6 subscales, the score of low-risk gamers on the loneliness subscale was significantly lower than those of non-gamers and high-risk gamers (non-gamers: 0.94 ± 1.44 , low-risk gamers: 0.53 ± 1.02 , and high-risk gamers: 1.02 ± 1.64 , $p=0.001$). On the other hand, there was no significant difference among the three groups on the social support ($p=0.839$) and the social network subscales ($p=0.055$). When only male respondents were targeted, low-risk gamers had the same tendency to have signif-

Table 2. Comparison of LSIS-6 and subscale scores according to the game usage

	(1) Non-gamers	(2) Low-risk gamers	(3) High-risk gamers	F	p	Post-hoc
All respondents	186	175	208			
LSIS-6	5.71 $\pm$ 3.19	4.97 $\pm$ 2.39	5.40 $\pm$ 3.22	2.84	0.059	
Loneliness	0.94 $\pm$ 1.44	0.53 $\pm$ 1.02	1.02 $\pm$ 1.64	6.68	0.001	(2)<(1), (3)
Social support	2.43 $\pm$ 1.68	2.36 $\pm$ 1.73	2.32 $\pm$ 1.84	0.18	0.839	
Social network	2.35 $\pm$ 1.44	2.09 $\pm$ 1.09	2.05 $\pm$ 1.37	2.92	0.055	
Male respondents	134	135	146			
LSIS-6	5.86 $\pm$ 3.30	4.86 $\pm$ 2.49	5.59 $\pm$ 3.17	4.12	0.017	(2)<(1), (3)
Loneliness	0.93 $\pm$ 1.49	0.48 $\pm$ 1.00	1.11 $\pm$ 1.64	7.32	<0.001	(2)<(1), (3)
Social support	2.44 $\pm$ 1.50	2.27 $\pm$ 1.69	2.39 $\pm$ 1.72	0.38	0.685	
Social network	2.49 $\pm$ 1.51	2.10 $\pm$ 1.11	2.09 $\pm$ 1.31	3.96	0.020	(1)>(2), (3)
Female respondents	52	39	62			
LSIS-6	5.33 $\pm$ 2.87	5.36 $\pm$ 2.34	4.93 $\pm$ 3.29	0.39	0.680	
Loneliness	0.96 $\pm$ 1.30	0.67 $\pm$ 1.09	0.82 $\pm$ 1.65	0.47	0.625	
Social support	2.39 $\pm$ 2.11	2.65 $\pm$ 1.80	2.16 $\pm$ 2.13	0.72	0.488	
Social network	1.99 $\pm$ 1.13	2.04 $\pm$ 1.06	1.95 $\pm$ 1.51	0.06	0.940	

Data are presented with number only or mean $\pm$ standard deviation. Post-hoc tests were performed with Bonferroni correction with p-value of <0.05. N, number of respondents; LSIS-6, Loneliness and Social Isolation Scale-6

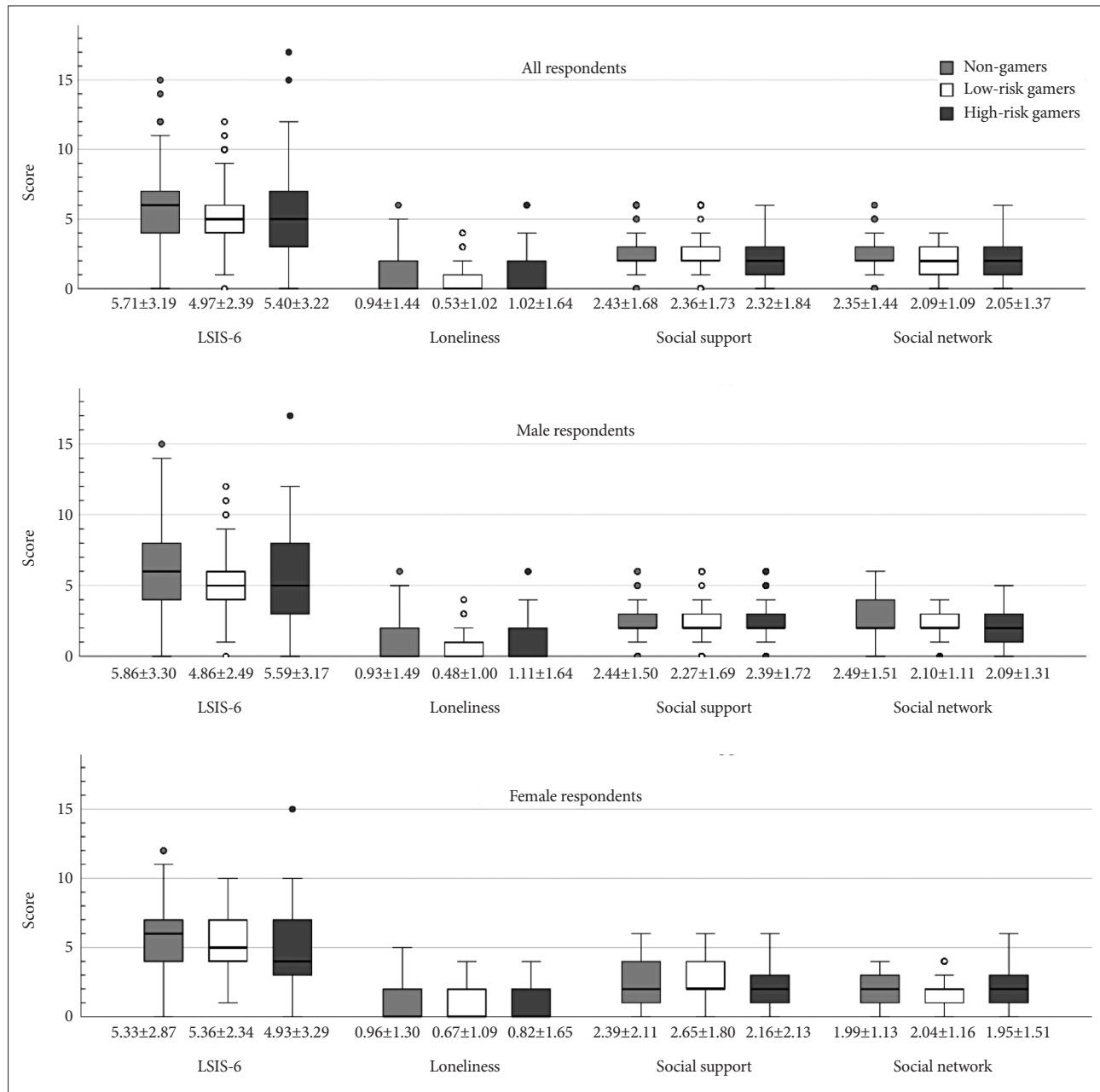


Figure 1. Comparison of LSIS-6 and subscale scores according to the game usage. LSIS-6, Loneliness and Social Isolation Scale-6.

icantly lower scores than the other two groups on the loneliness subscale (non-gamers: 0.93 ± 1.49 , low-risk gamers: 0.48 ± 1.00 , and high-risk gamers: 1.11 ± 1.64 , $p < 0.001$). In addition, in male respondents, the score of non-gamers on the social network subscale was higher than those of low-risk or high-risk gamers, and this difference was statistically significant (non-gamers: 2.49 ± 1.51 , low-risk gamers: 2.10 ± 1.11 , and high-risk gamers: 2.09 ± 1.31 , $p = 0.020$). Even among male respondents, there was no significant difference among the three groups in the scores for the social support subscale ($p = 0.685$). In the subscale analyses of female respondents, all three sub-scales

showed no significant differences in scores among the three game usage groups.

Multivariate linear regression models for LSIS-6 with the game usage

Among the sociodemographic factors, employment status was excluded from the multivariate linear regression model because it showed a strong correlation with sex (χ^2 , $p < 0.001$). Table 3 shows multivariate linear regression models for LSIS-6 and each of its subscales. When the score of low-risk gamers is used as a reference, it was found that non-game usage was

Table 3. Association between LSIS-6 and its subscales with the game usage

	LSIS-6				Loneliness subscale				Social support subscale				Social network subscale			
	B	S.E.	t	p	B	S.E.	t	p	B	S.E.	t	p	B	S.E.	t	p
Non-gamers	0.74	0.30	2.43	0.016	0.41	0.14	2.94	0.003	0.06	0.18	0.31	0.760	0.27	0.14	1.95	0.052
Low-risk gamers	Ref.				Ref.				Ref.				Ref.			
High-risk gamers	0.38	0.29	1.29	0.197	0.44	0.14	3.18	0.002	-0.02	0.18	-0.12	0.905	-0.03	0.13	-0.23	0.817

In each model, sex, age group, marital status, years of education, depressive disorder, anxiety disorder, and substance use disorder were adjusted for. B, unstandardized coefficient; S.E., standard error; t, t-statistics; LSIS-6, Loneliness and Social Isolation Scale-6

statistically significantly associated with an increased overall LSIS-6 score ($B=0.74$, $t=2.43$, $p=0.016$). On the other hand, the regression coefficient of being a high-risk gamer in LSIS-6 was not significant ($p=0.197$). In the model for the loneliness subscale, both non-gamers ($B=0.41$, $t=2.94$, $p=0.003$) and high-risk gamers ($B=0.44$, $t=3.18$, $p=0.002$) were found to have statistically significantly higher coefficients when compared against the reference (low-risk gamers). In the model for the social support subscale, the association of game usage group and social support subscale score was not significant (non-gamers: $p=0.760$ and high-risk gamers: $p=0.905$). In the model for the social network subscale, it was also found that neither the non-gamers ($p=0.052$) nor the high-risk gamers ($p=0.817$) had a statistically significant difference from the reference value, the low-risk gamers.

DISCUSSION

This is the first study to explore the associations among loneliness, social isolation, and game use adjusting for confounding effects of psychiatric disease in a large-scale census. Our findings differ from those of previous studies in those psychiatric disorders such as depressive disorder, anxiety disorder, or substance disorder were given consideration, and in that a large amount of data was used to represent the general population.²⁷

When confounding variables including psychiatric diseases were corrected for, low-risk gamers showed lower scores for loneliness than non-gamers and high-risk gamers. Also, in the case of male sex, low-risk gamers showed statistically significantly lower scores for loneliness and social network. When reviewing previous study results significantly correlating social isolation with loneliness,¹⁹ our results mean that male low-risk gamers today have more social interaction, are more socially connected, and feel less lonely. This finding correlates with previous findings¹¹ that Internet game stimuli correlate with social stimuli in brain studies. The results of this study also corresponded with those of previous studies in that loneliness and social isolation had similar aspects and correlations but also had different characteristics. In previous studies,²⁸ loneliness was more frequent in females, while so-

cial isolation was more frequent in males. Also, no statistically significant age difference was observed for loneliness, but social isolation was more prevalent in ages 30–44 and 60–74 years.²⁸ In this study, different patterns of loneliness and social isolation were also observed between genders.

The study showed that female gamers showed no significant difference in loneliness and social isolation. A previous study has shown that females feel loneliness and isolation more easily than do males,²⁹ and another study showed females are more resistant to gaming use disorder and have high resilience to game addiction.³⁰ A previous study found different game play styles between males and females,³¹ and this difference may alter the effects of game use on loneliness and social connectivity. Accordingly, that game use acts differently as a protective factor against loneliness and social isolation between male and female gamers owes to differences in how males and females experience stimuli such as loneliness, social isolation, and gaming. The present study had fewer female gamer respondents than males, and further studies are needed to evaluate this finding for a sufficient number of female subjects.

A strength of the present study is that samples were collected nationwide. This is the first study to explore the relationships among loneliness, social isolation, and online game use using mental disorder data from a wide-ranging population census, and its findings may be more representative of the general population.²⁷ Second, given the many social changes since the COVID-19 pandemic, the topics of the present study—online games, loneliness, and social isolation—have been explored, but not enough information has been amassed thereon to date. The results of our study can be of greater significance in this aspect. Third, our results show that online gaming is associated with protection from loneliness and social isolation. This can be a meaningful finding among the social discourse with regard to the effects of online gaming.^{32,33} The findings of this investigation are consistent with recent research which suggests that use of virtual reality can positively affect emotion, and that interpersonal connections can be reinforced online.³⁴ Fourth, propensity score matching was conducted in our study, potentially yielding more statistically strict result with large-scale demographic data.

The present study also had several limitations. First, many recent studies studying loneliness, social isolation, and gaming use have employed different measurement tools,³⁵⁻³⁸ and many self-reported scales were revised in the process. This may have lowered the representativeness and validity of the present study due to methodological issues. Furthermore, the distinction between single-player and multiplayer online games is crucial for understanding gaming-related social interactions.^{39,40} While this study's scale did not capture these differences, future research should explore their impact on loneliness and social isolation. Second, our results were drawn from a limited sample population. Whereas the sample size of the study was large, all samples were collected from South Korea. Generalizability is not warranted in different socio-demographic situations, and further studies focused on different ethnicities or nationalities are needed. Further, the study being a cross-sectional observation study, interpretation of temporal relationships among the traits dealt with herein is not warranted. Therefore, longitudinal research is necessary to supplement the limitations of this study. Additionally, while the high-risk gamer group is typically male-dominated,⁴¹ propensity score matching in this study led to a relatively higher proportion of high-risk female gamers compared to the original population. Therefore, the observed gender differences may not reflect the actual population, requiring further validation in future research. Finally, the study population was recruited during the COVID-19 pandemic, and this may have acted as a confounding variable. Increased mental health risks such as depression, anxiety, and stress have been reported since the COVID-19 pandemic,^{42,43} and negative effects of reduced physical activity and social connection on mental health after the pandemic have also been reported.^{44,45} Moreover, previous studies have suggested that online gaming use has increased as a coping strategy during the pandemic.⁴⁶ This raises the possibility that individuals with higher levels of loneliness may engage in online gaming more frequently, introducing a potential confounding effect. Therefore, caution is needed when interpreting the findings of this study, and further research is required to explore the long-term impact of pandemic-related gaming behaviors on loneliness and social interactions.

In conclusion, low-risk game use was associated with low levels of loneliness and social isolation in males. When correcting for the confounding effects of variables such as mental health problems, the results of the present study showed the effect of game use on loneliness and social isolation was not statistically significant among high-risk gamers. Rather, gaming was found to reduce loneliness and social isolation among low-risk gamers. Further studies are needed to determine the specific reasons why this may be, and develop methodological approaches to apply our results in determining the types

of game and duration of game use that may yield such beneficial effects.

Supplementary Materials

The Supplement is available with this article at <https://doi.org/10.30773/pi.2023.0385>.

Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author on reasonable request.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

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